

# SUPPLEMENTARY

## Supplementary Material — Centrality-Based KV-Cache Eviction

Companion to the main report (REPORT.pdf): detailed per-press systems figures and per-task result tables. See the report §4.4–4.5 for the summary and interpretation.

### S.1 Per-press systems figures

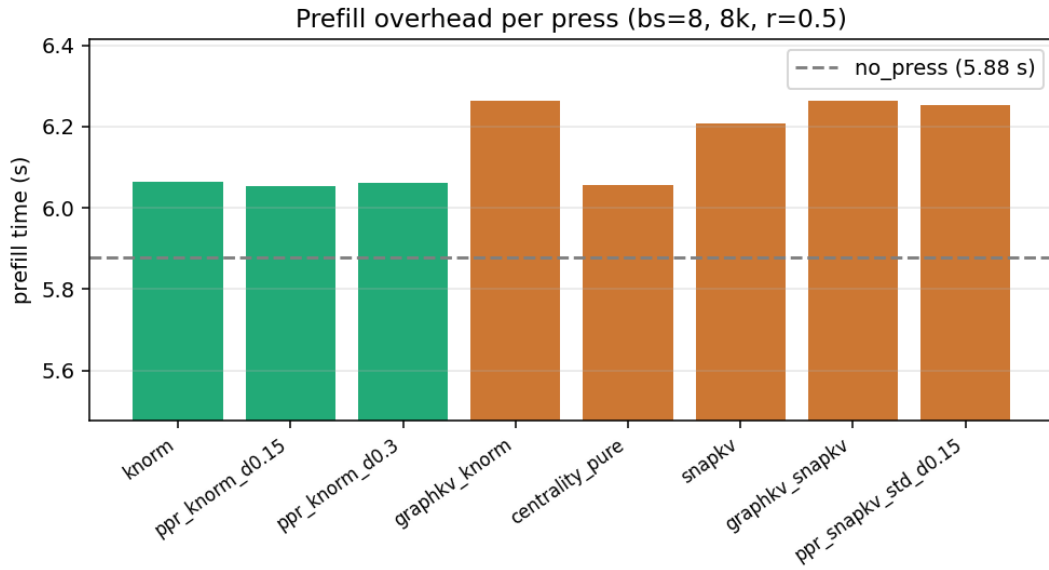


Figure S1: Prefill scoring overhead — the only press-dependent systems cost;  $ppr\_knorm \approx knorm$ , cheaper than  $graphkv$  /  $SnapKV$ -based presses.

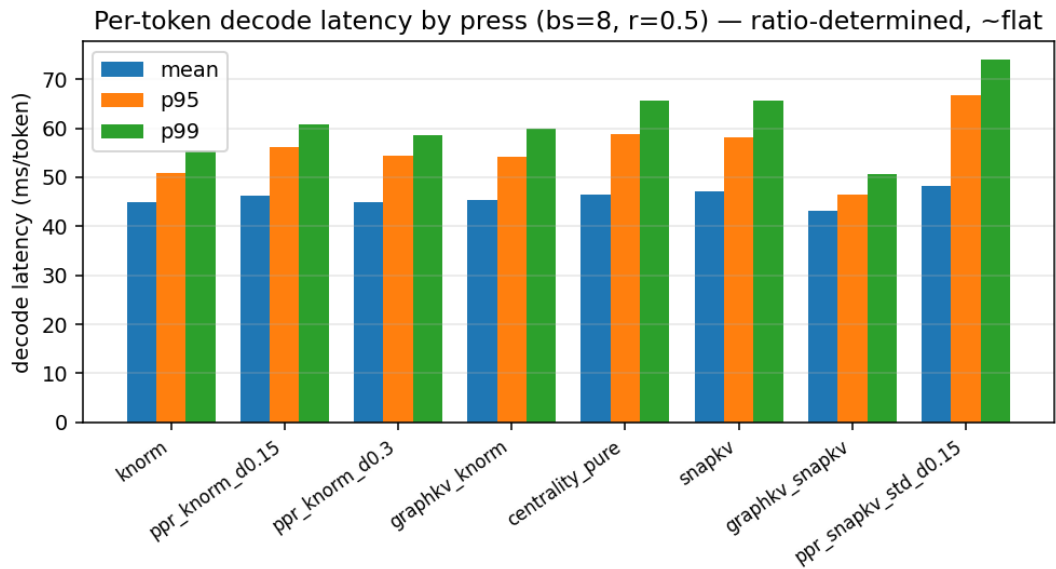


Figure S2: Per-token decode latency (mean/p95/p99) by press at  $r=0.5$  — ratio-determined, ~flat.

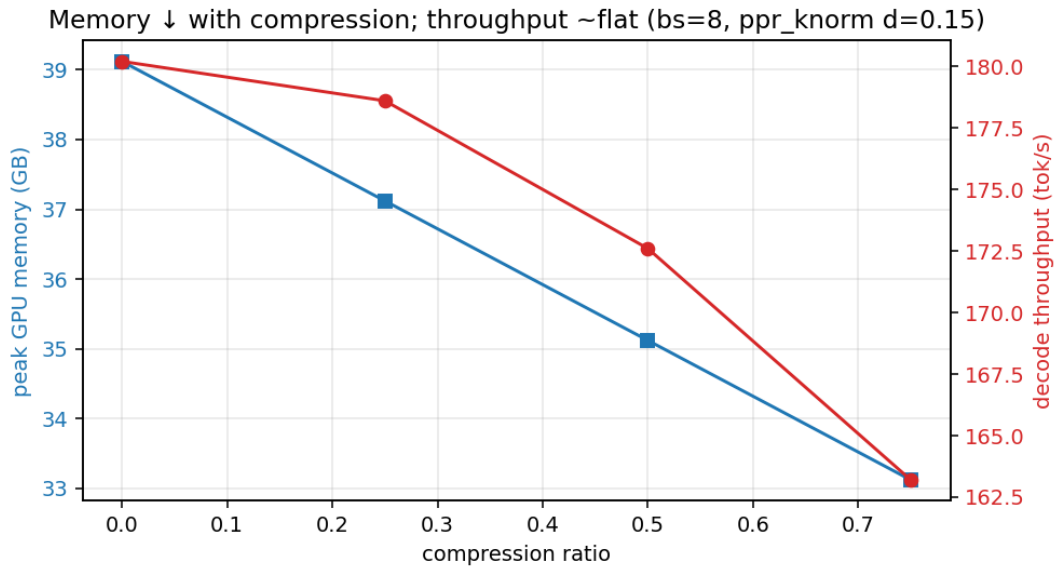


Figure S3: Peak GPU memory falls linearly with compression; decode throughput ~flat at this scale.

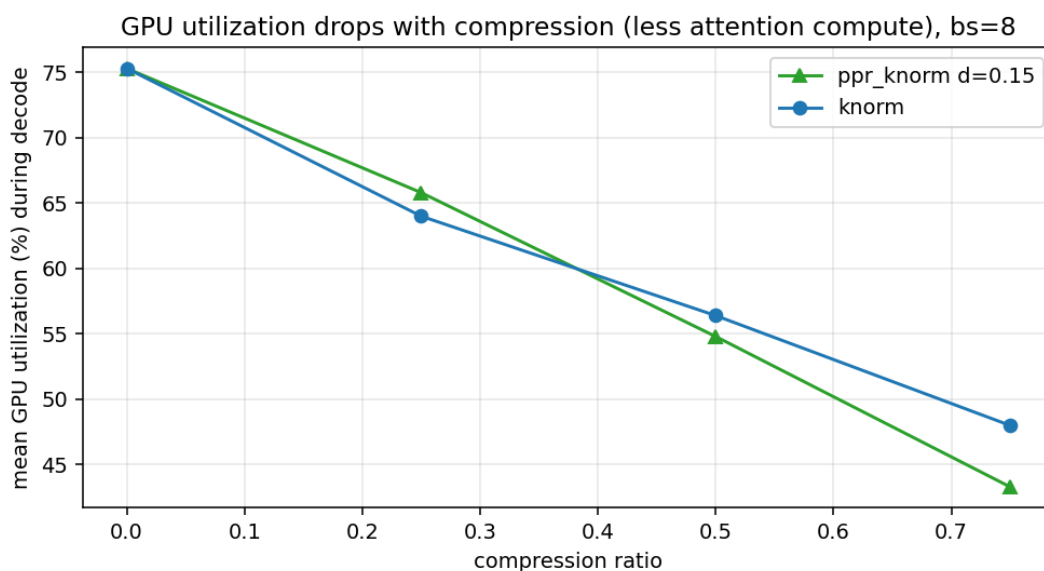


Figure S4: Mean decode GPU utilization drops with compression (less attention compute per token).

## S.2 Per-task tables

### RULER 4096 @ compression 0.5 (string-match %, usable tasks)

| task            | no_press | knorm | graphkv | snapkv | ppr d=0.15 | pure |
|-----------------|----------|-------|---------|--------|------------|------|
| niah_single_1   | 100      | 100   | 100     | 94.7   | 100        | 0    |
| niah_single_2   | 100      | 80.0  | 20.0    | 97.1   | 100        | 0    |
| niah_single_3   | 100      | 2.9   | 5.9     | 0      | 73.5       | 0    |
| niah_multikey_1 | 100      | 30.0  | 40.0    | 90.0   | 90.0       | 6.7  |
| niah_multikey_2 | 100      | 4.8   | 81.0    | 61.9   | 100        | 0    |
| niah_multikey_3 | 100      | 4.5   | 36.4    | 22.7   | 18.2       | 0    |
| qa_2            | 58.1     | 32.3  | 48.4    | 48.4   | 48.4       | 29.0 |

### LongBench multi-hop @ compression 0.5 (F1 %)

| subtask  | no_press | knorm | graphkv | snapkv | ppr d=0.15 |
|----------|----------|-------|---------|--------|------------|
| hotpotqa | 58.5     | 54.1  | 45.9    | 56.7   | 49.2       |
| 2wikimqa | 53.1     | 45.9  | 46.7    | 53.5   | 53.4       |
| musique  | 30.7     | 19.5  | 30.9    | 25.2   | 23.4       |

Per-run configs + metrics are under Appendix/results/ (config.yaml + metrics.json each; the aggregated per-example \*\_long.csv are report-only — regenerable, see Appendix/MANIFEST.md). Regenerate figures with analysis/make\_figures.py and paired stats with analysis/analyze\_paired.py.

## S.3 Iso-accuracy operating-point curve (16k, batch 8)

16k / batch 8: memory scales linearly with cache; speed moves only past the knee

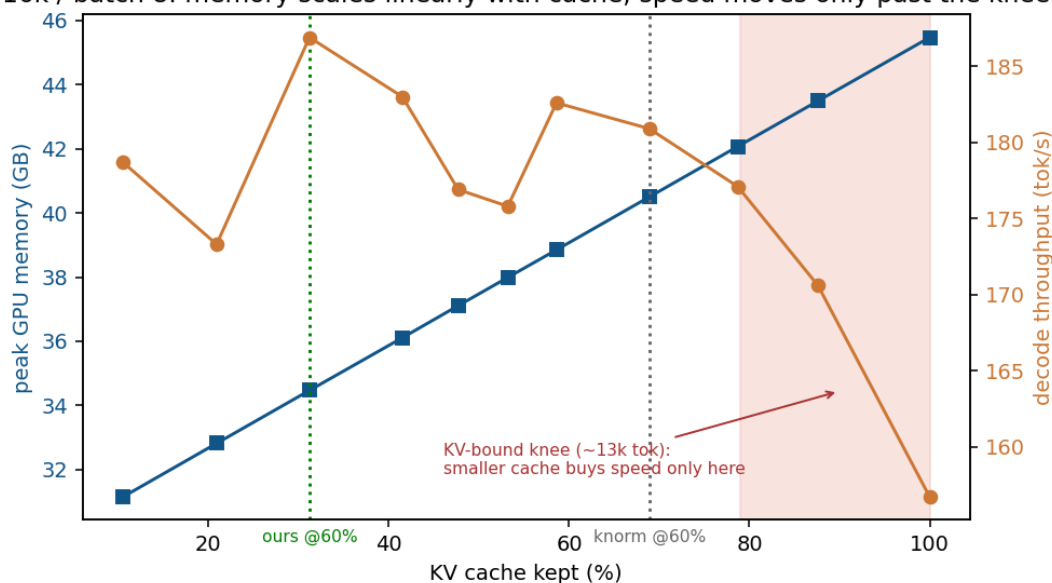


Figure S5: Directly measured peak GPU memory and decode throughput vs. kept-cache fraction (16k prompt, batch 8, analysis/iso\_systems.py on results/results\_iso\_systems\_16k\_bs8.csv). Peak memory falls linearly with the cache; throughput is essentially flat once the cache is compressed (decode is weight-bound at 8B/16k/bs8). The dotted lines mark the iso-accuracy operating points for a 60 % RULER target: the base press must keep ~69 % of the cache, ours only ~31 % — same accuracy,  $2.2\times$  less cache and ~6 GB less peak memory, while throughput is unchanged (the speed win needs a more KV-bound regime).

## Appendix — Code & data artifacts

- **Press code:** kvpress/presses/centrality\_press.py; registration in kvpress/\_\_init\_\_.py, evaluation/evaluate\_registry.py, tests/default\_presses.py, README.md; unit test in tests/presses/test\_centrality\_press.py.
- **GraphKV comparison baseline (additional\_benchmarks/):** decay\_propagation\_press.py, test\_decay\_propagation\_press.py, run.py (runtime-injected registry entry), README.md.
- **Benchmark + analysis (Appendix/):** scripts/ruler\_rerun.py, scripts/sweep\_longbench.py, scripts/tau\_sweep.sh, scripts/submission\_grid\_llama.sh, scripts/submission\_grid\_qwen.sh, scripts/fetch\_eval\_datasets.sh, scripts/eval\_cfg\_dec.yaml; analysis/analyze\_paired.py, analysis/make\_figures.py, analysis/speed\_memory.py, analysis/iso\_systems.py, analysis/recall\_probe.py, analysis/rank\_vs\_board.py, analysis/teleport\_entropy.py, analysis/kvpress\_leaderboard\_raw.csv.
- **Data/logs (Appendix/results/):** board\_grid/, ruler\_screen/, and ablations/ run directories (each with config.yaml + metrics.json; large predictions.csv omitted); figures in report/figures/ (fig1–fig12).

- **Reproducibility:** requirements.txt, Dockerfile, README.md (how-to-benchmark), MANIFEST.md (report number → run dir), BASELINE.md, REGENERATE.md, .github/workflows/centrality-ci.yml (CI: comparison-benchmark tests on every push).